

Predictive Maintenance for Industrial Equipment Using Artificial Intelligence

B.Tech CSE (AI & ML) Mini Project Report

Submitted By: Rashi Goel

College: Dronacharya College of Engineering

Certificate

This is to certify that this project has been completed in partial fulfillment of the B.Tech degree requirements.

Acknowledgement

I thank my guide, faculty members, friends and family for their support.

Abstract

AI-based predictive maintenance predicts industrial equipment failures using sensor data and a Random Forest model. The project includes preprocessing, training, real-time monitoring and a Streamlit dashboard.

Introduction

Predictive maintenance uses machine learning to identify possible failures before breakdowns occur, reducing downtime and maintenance cost.

Problem Statement & Objectives

Develop a predictive maintenance model, monitor equipment in real time, generate alerts and visualize equipment health.

Methodology

Collect data → Preprocess → Train Random Forest → Predict → Alert → Dashboard.

Requirements

Python, Pandas, NumPy, Scikit-learn, Joblib, Streamlit, VS Code.

Results

The model classifies equipment as Healthy or Failure and recommends maintenance actions.

Advantages

Reduced downtime, early fault detection, improved productivity.

Future Scope

IoT integration, cloud deployment, mobile app, deep learning, email alerts.

Conclusion

The project demonstrates AI-enabled predictive maintenance for industrial equipment.

References

Scikit-learn, Python, Streamlit documentation; Hands-On Machine Learning.